Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal line” and “reflection” to explain your answer. Sentence starters are optional.

normal line: a reference line perpendicular to a surface

reflection: a wave returning from a boundary

virtual image: an image at an apparent origin of rays, not their meeting point

01 Warm-up: bring back an earlier idea

A wall is **50 m** away. An echo returns after **0.5 s**. A student divides one-way distance by this time. **Find the correct speed** and **explain the student’s error**.

Word help: echo = a sound received after reflection

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: ① ② ③

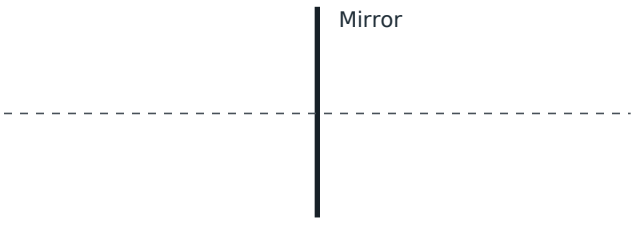
02 Worked example: explain the prediction

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; **measure angles** from the normal. The image is virtual.

Worked reasoning: An object **2 m** in front of an ideal plane mirror has a virtual image **2 m** behind it. The object-image separation is **4 m**; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

Sentence starter: In my work, I used … to show … . I labeled it … .



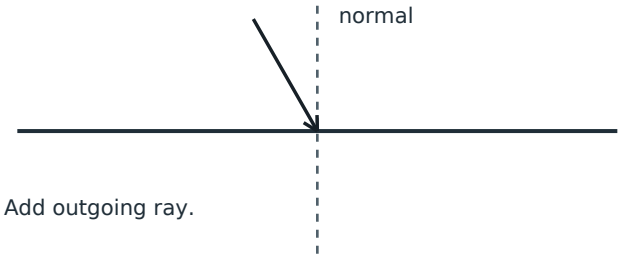
Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw the normal at the mirror. **Complete a reflected ray** for an incident ray **30°** from normal.

Sentence starter: In my work, I used … to show … . I labeled it … .



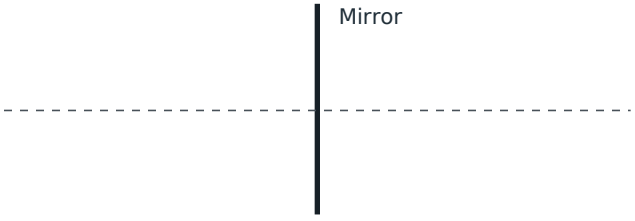
Add outgoing ray.

Marzano check - circle: ① ② ③

04 Apply the physics

A small lamp is **2 m** in front of a plane mirror. **Mark the lamp** and its virtual image on the diagram. **Find image distance behind the** mirror and the full lamp-to-image separation.

Sentence starter: My design includes … . It does what is needed by … .



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is the mirror image called virtual rather than a real object behind the glass?

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A reflected signal returns after **4 s**. **Name the extra quantity needed** to **calculate distance** and **explain why this** is a round trip.
Label the outgoing and returning paths.

Word help: signal = a disturbance that carries information

Sentence starter: I think … . I can point to …, which shows …, so … .

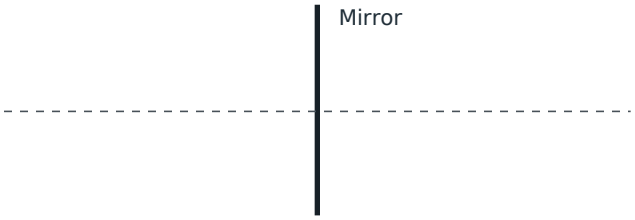


Marzano check - circle: 1 2 3

07 Exit check

Incidence angle **45°** from normal. **Find reflection angle**.

Sentence starter: I found … [unit]. This value means … in this situation.



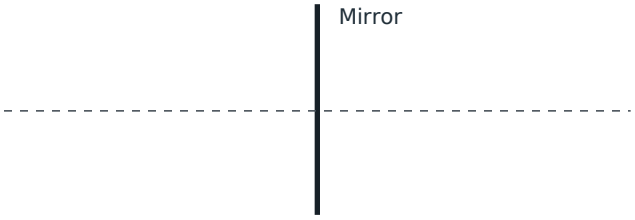
Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Move the object **0.5 m** closer to the mirror. **Find the new image distance** and object-image separation.

Sentence starter: I found … [unit]. This value means … in this situation.



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3